

1. Student Details

Name	Md Faiz Alam
Course	BCA (Bachelor of Computer Applications)
Semester	5 th Semester
Admission No.	UGO24156735
GitHub	https://github.com/Md-Faiz-Alam/quizmind-ai
Live App	https://quizmind-faiz-alam-projects-a349ef8e.vercel.app/
Backend API	https://quizmind-ai-t6xc.onrender.com

2. Project Title & Objective

Title: QuizMind AI – An Adaptive Quiz Platform for Interview & Placement Preparation

Objective: QuizMind AI is a fully deployed web-based adaptive quiz platform for computer science and data science students preparing for technical interviews and campus placements. The platform provides 894 real interview-style MCQ questions across 6 major topics and 35 subtopics, tracks user performance at the subtopic level, detects weak areas, and provides instant explanations after every wrong answer — all accessible freely from any device anywhere in the world.

3. Final Features Implemented

- User registration and secure login with SHA-256 password hashing.
- Dashboard showing all 6 topics and 35 subtopics with performance progress bars.
- Adaptive quiz engine with 894 real interview-style MCQ questions.
- 30-second countdown timer per question with auto-progression.
- Instant explanation panel shown after each wrong answer.
- Color-coded answer highlighting — green for correct, red for wrong.
- Results page with score, accuracy percentage, correct/wrong count, and time taken.
- AI Insight panel giving personalized study suggestion based on quiz performance.
- Analytics dashboard with bar chart of accuracy by topic using Recharts.
- Weak area detection — automatically identifies subtopics below 60% accuracy.
- Recent quiz history tracking across all attempts.
- Cloud PostgreSQL database on Supabase — data persists permanently.
- Flask REST API backend with authentication, quiz and result routes.
- Fully deployed — frontend on Vercel, backend on Render.com.
- Accessible from any device and any location worldwide.

4. System Working Explanation

The system works in the following steps:

- **Step 1 — Register / Login:** User visits the live URL, registers with name, email and password or logs into an existing account. Password is hashed using SHA-256 before storing in Supabase.
- **Step 2 — Dashboard:** After login, the dashboard loads all 6 topics (Python, SQL, DSA, Machine Learning, Statistics, Data Analysis) with their subtopics fetched from the cloud database via Flask API.
- **Step 3 — Select Subtopic:** User clicks any subtopic button. The quiz screen loads questions for that subtopic from the Flask API which queries Supabase.
- **Step 4 — Attempt Quiz:** Each question is displayed with 4 options and a 30-second countdown timer. User selects an answer — correct option turns green, wrong option turns red, and explanation is shown instantly.
- **Step 5 — Save Results:** After completing all questions, the result (score, accuracy, time taken) is automatically saved to Supabase via a POST API call.
- **Step 6 — View Results:** Results page shows score, accuracy percentage, correct/wrong breakdown, time taken, and an AI-generated study insight.
- **Step 7 — Analytics:** Analytics page fetches all past results from Supabase, displays accuracy by topic as a bar chart, identifies weak areas below 60% accuracy, and shows quiz history.

5. Conclusion

QuizMind AI was successfully built and deployed as a fully functional web application. The project achieves its core objective of providing an intelligent, free, and accessible quiz platform for students preparing for technical interviews and campus placements.

What was achieved:

- Built a complete full-stack web application from scratch using React.js and Flask
- Created a cloud-hosted database with 894 real interview questions across 35 subtopics
- Implemented user authentication, quiz engine, result tracking, and analytics
- Successfully deployed the application — live and accessible from anywhere in the world
- Learned full-stack development, REST API design, cloud database management, and deployment
- Learned importance of separating concerns — frontend, backend, and database layers. Git and GitHub for version control and collaboration.